

News from custom code migration

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Custom code migration in 2025 and Q1/2026

| SAP Joule for Developers ABAP AI capabilities in 2025

[Generative AI in ABAP Cloud](#)

Accelerate

Joule for developers

Increase developer productivity
Integrated in ABAP development tools

Empower

ABAP AI SDK powered by ISLM

Empower developers to infuse AI into business scenarios
ABAP client for the generative AI hub in the SAP AI Core infrastructure

Transform

ECC → SAP S/4HANA

Custom Code Migration AI assistant integrated in the ABAP test cockpit
and the Custom Code Migration app (since Q1 2026)

/ Learn more: [Custom Code Migration to SAP S/4HANA powered by SAP Joule for Developers, ABAP AI capabilities](#)

Custom code migration with ABAP AI in 2025 - Q1/2026



AI-based support for IT project manager

Ai-based support for remote code analysis of ECC systems for S/4HANA

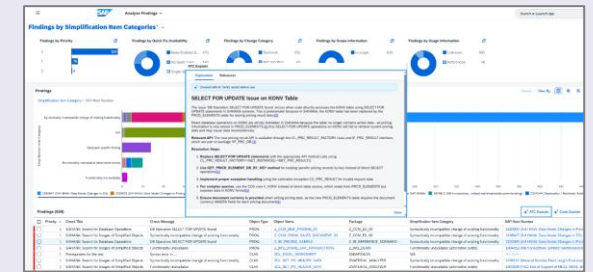
Embedded in Custom Code Migration app in SAP BTP ABAP environment

Explain of legacy code (reports and classes)

Explain of S/4HANA related ABAP test cockpit findings



Custom Code Migration app on SAP BTP



AI-based support for developer / functional consultant

Provided with SAP S/4HANA Cloud Private Edition 2025

Embedded in ABAP development tools for Eclipse (Joule copilot)

Docs chat for custom code migration to SAP S/4HANA

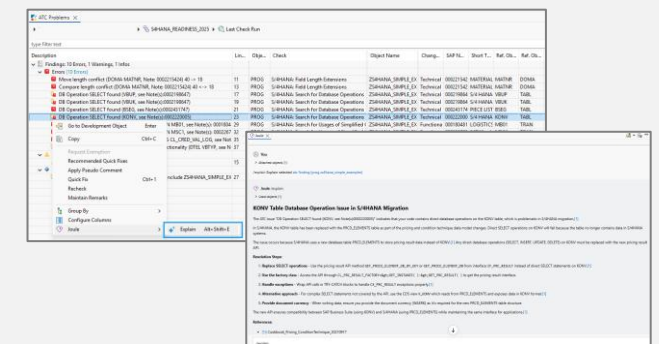
Explain legacy code (reports)

Explain S/4HANA related ABAP test cockpit findings

Explain for classes and module pools, code proposal for adaptation (with SAP S/4HANA Cloud Private Edition 2025 FPS01)



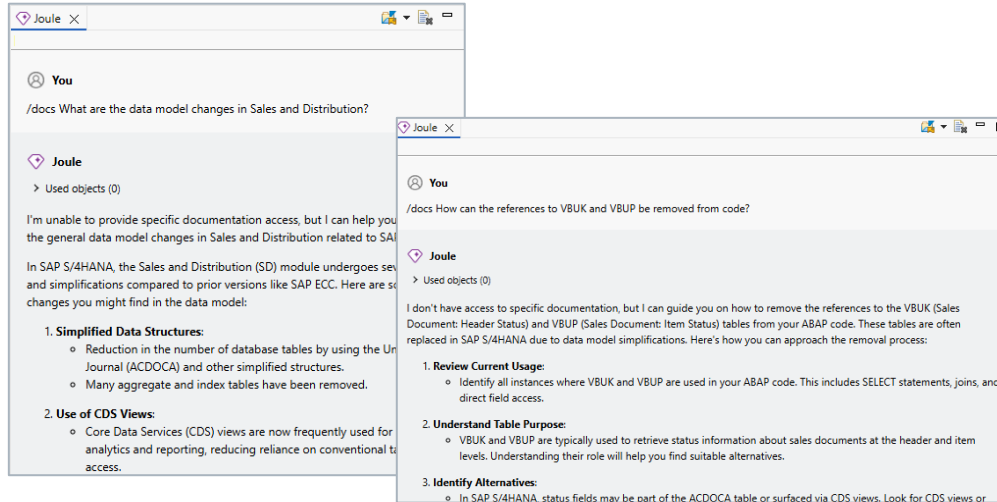
ABAP development tools for Eclipse



/ Learn more: [Custom code migration to SAP S/4HANA powered by SAP Joule for Developers, ABAP AI capabilities](#)

Custom code analysis and adaptation for SAP S/4HANA 2025

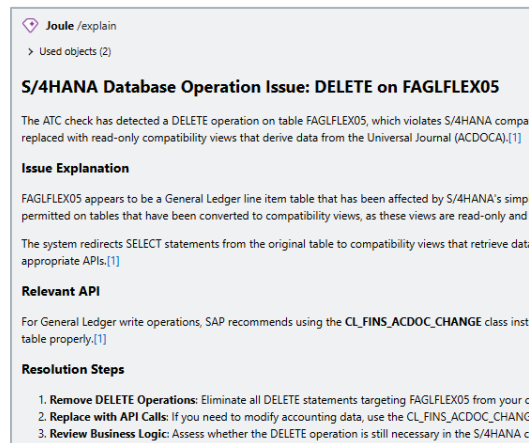
 Use `/docs chat` to get quick and accurate answers to questions around SAP S/4HANA simplification impact on custom code



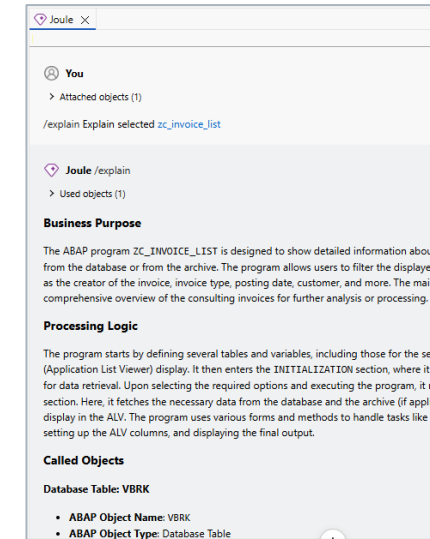
See also: [Docs chat video on YouTube](#)

 Explain ABAP test cockpit finding to quickly grasp the root cause of the SAP S/4HANA–related issue in your custom code

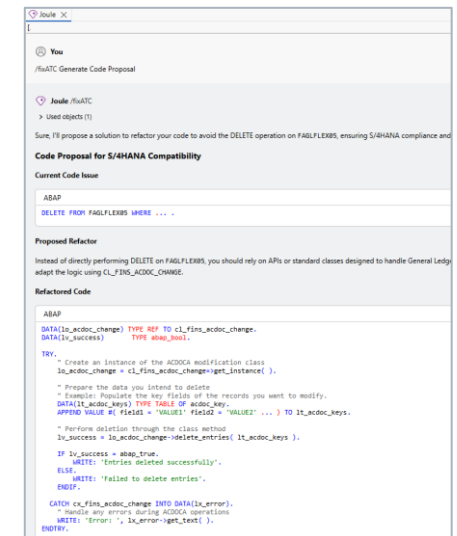
See also: [Explain ATC video on YouTube](#)



 Explain report to quickly understand business purpose and processing logic of the legacy custom code, which must be adapted to SAP S/4HANA



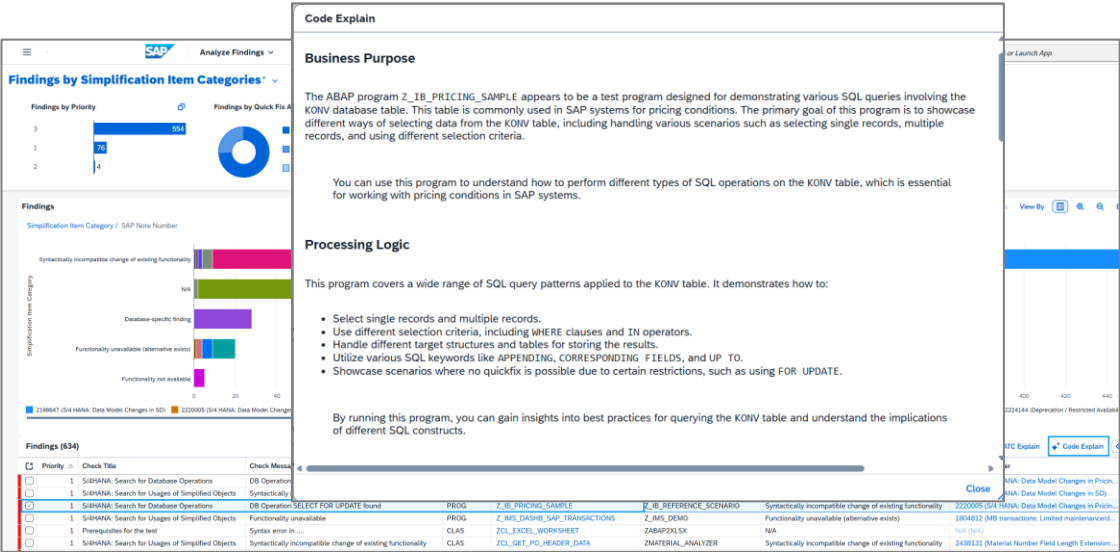
 Use code proposal to quick-fix the SAP S/4HANA–related issue in your custom code



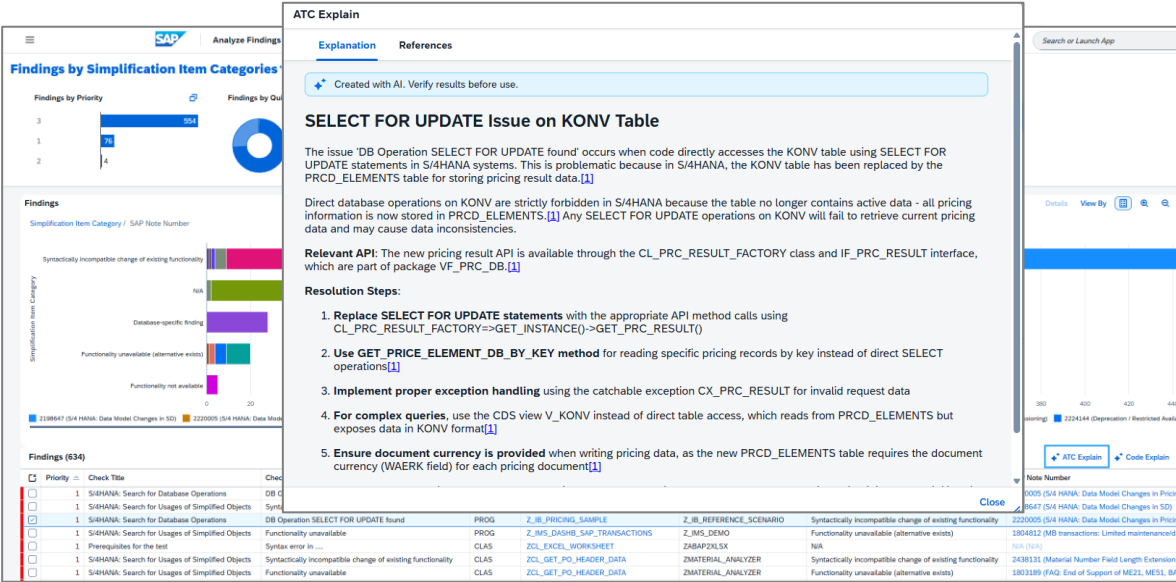
AI-based custom code analysis in the Custom Code Migration app

For remote analysis of ECC systems during conversion to SAP S/4HANA

Explain of legacy code (reports, classes)



Explain of S/4HANA related ABAP test cockpit findings



Custom code migration in 2026

Custom code adaptation for SAP S/4HANA on ECC system

Adapt your code for S/4HANA using ECC quick fixes

Analyze your custom code in ECC system with Custom Code Migration app on SAP BTP

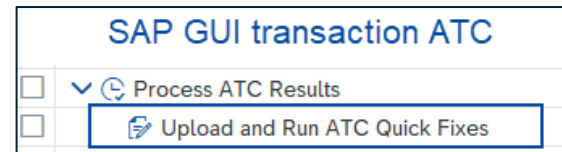
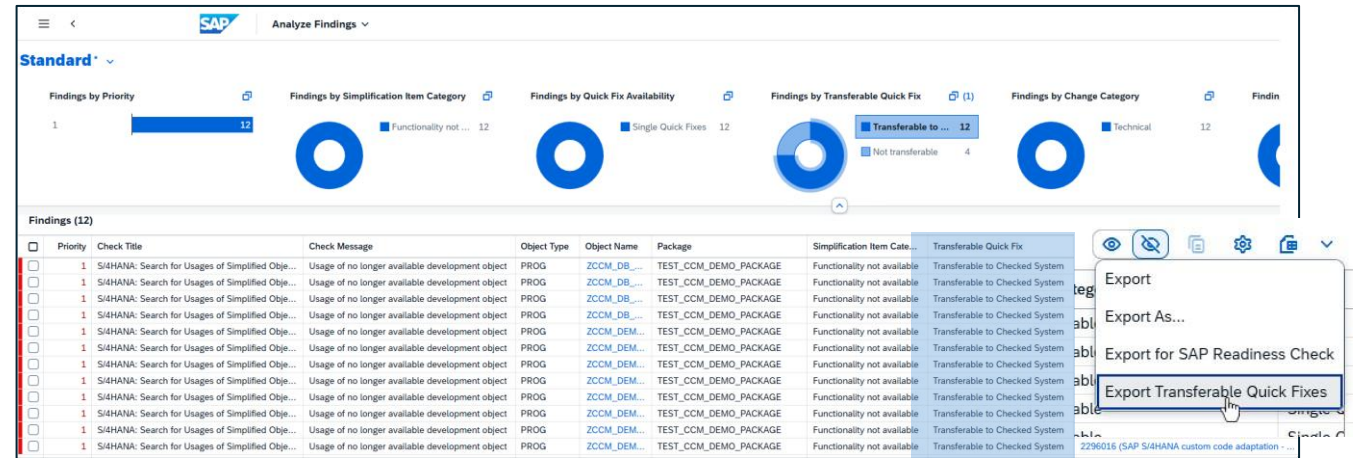
Download transferable quick fixes

Upload and run transferable quick fixes on your ECC system

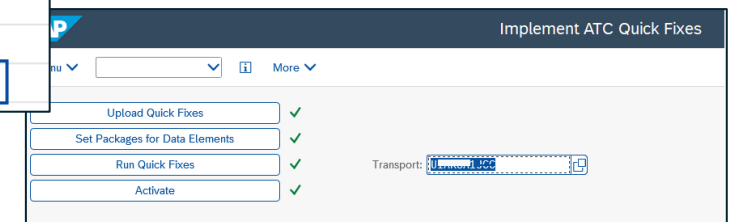
Transferable quick fixes are available

- For removed SAP standard Data Elements
- For missing ORDER BY in SELECT statements

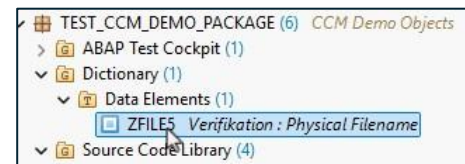
Download transferable quick fixes from Custom Code Migration app



Upload and run on ECC system



Custom data elements are created



Custom Code Analysis Projects

UI Plugin Sample

SAP

Custom Code Analysis Projects

Search or Launch App

OD

Standard

2 filters active: Project Type, Project Created By

Projects (1)

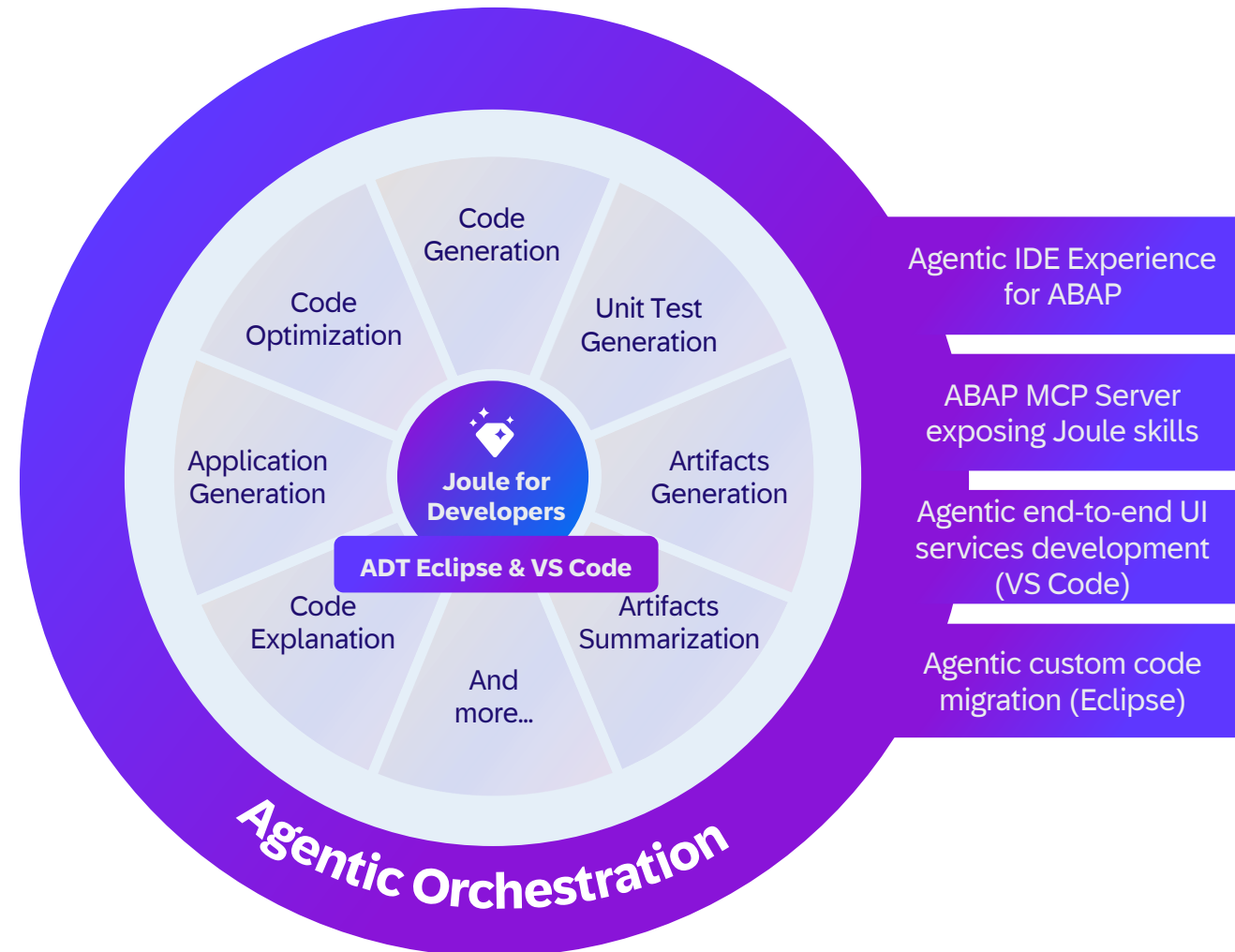
Create

Delete

Description	Project Type	Project Created At	Analysis State	Number of Findings	
My SAP S/4HANA custom code migration project	SAP S/4HANA migration	22.04.2026, 13:41:19	Finished	16	>

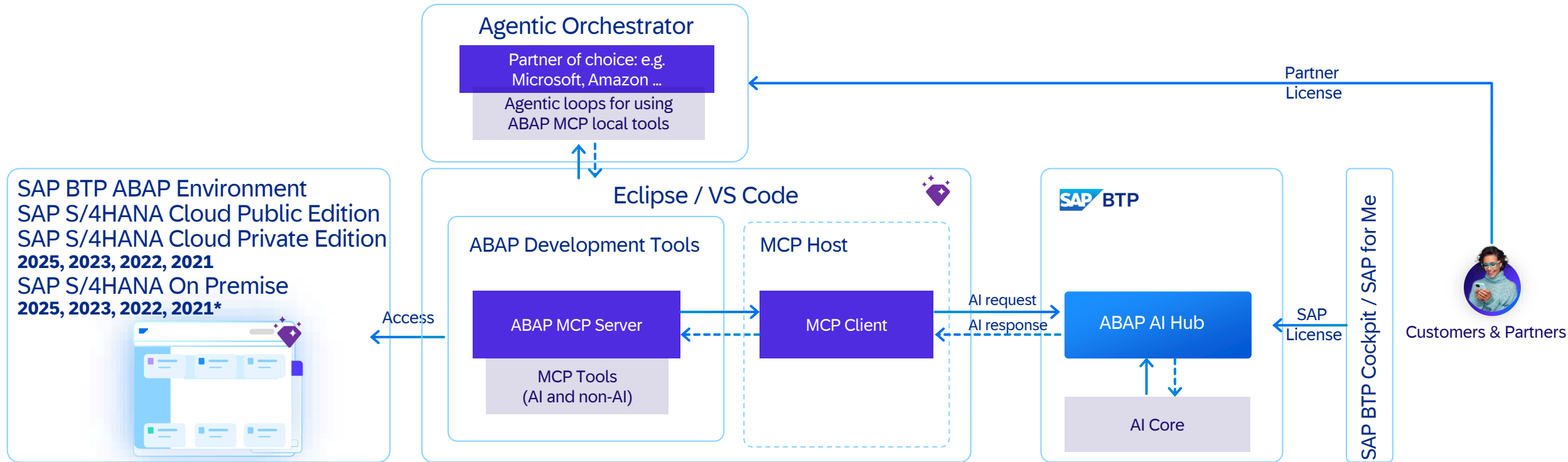
SAP Joule for Developers ABAP Agentic AI in 2026

Agent support in **VS Code and Eclipse** with **3rd party orchestrator** and **SAP ABAP MCP server delivered in Q2/2026**



SAP Joule for Developers ABAP Agentic AI

Side-by-side architecture as a foundation for all ABAP AI capabilities



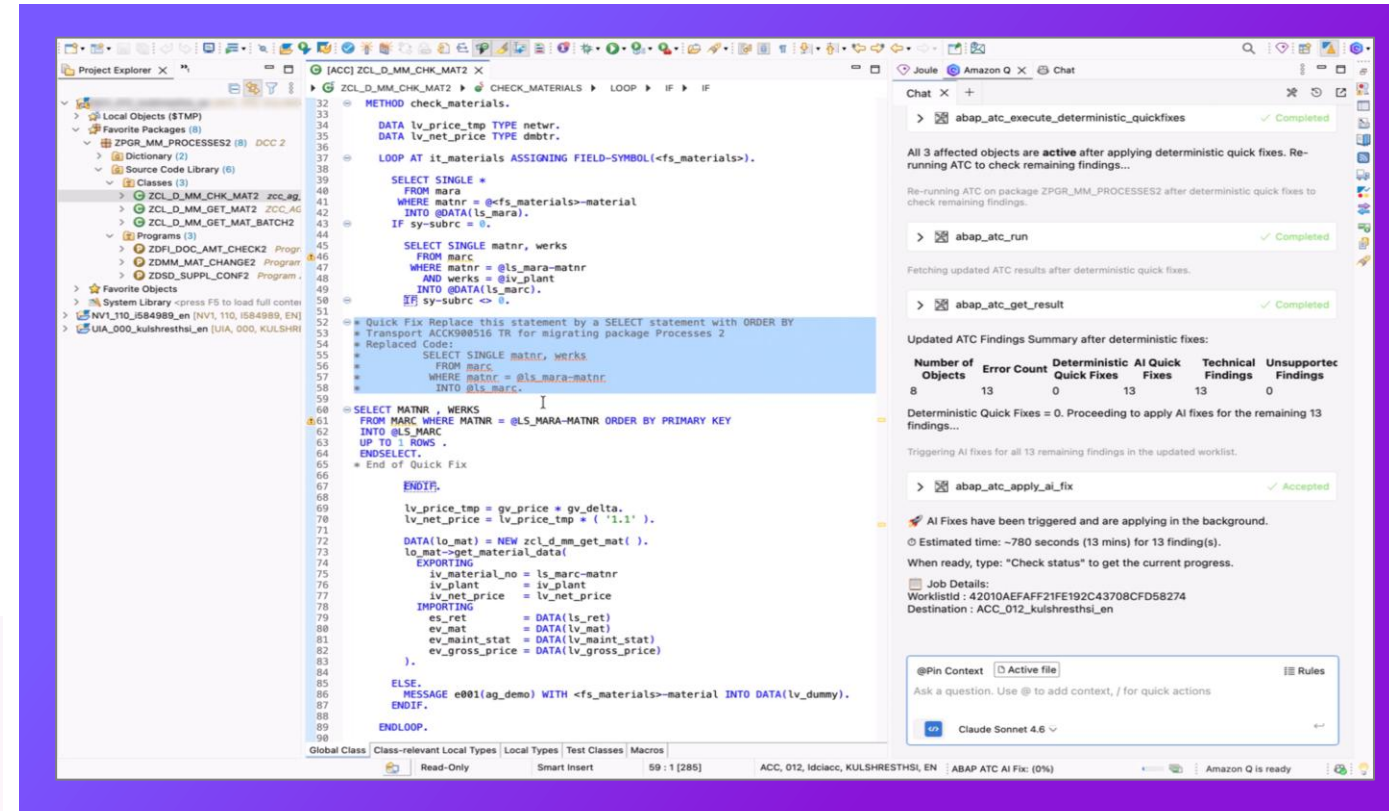
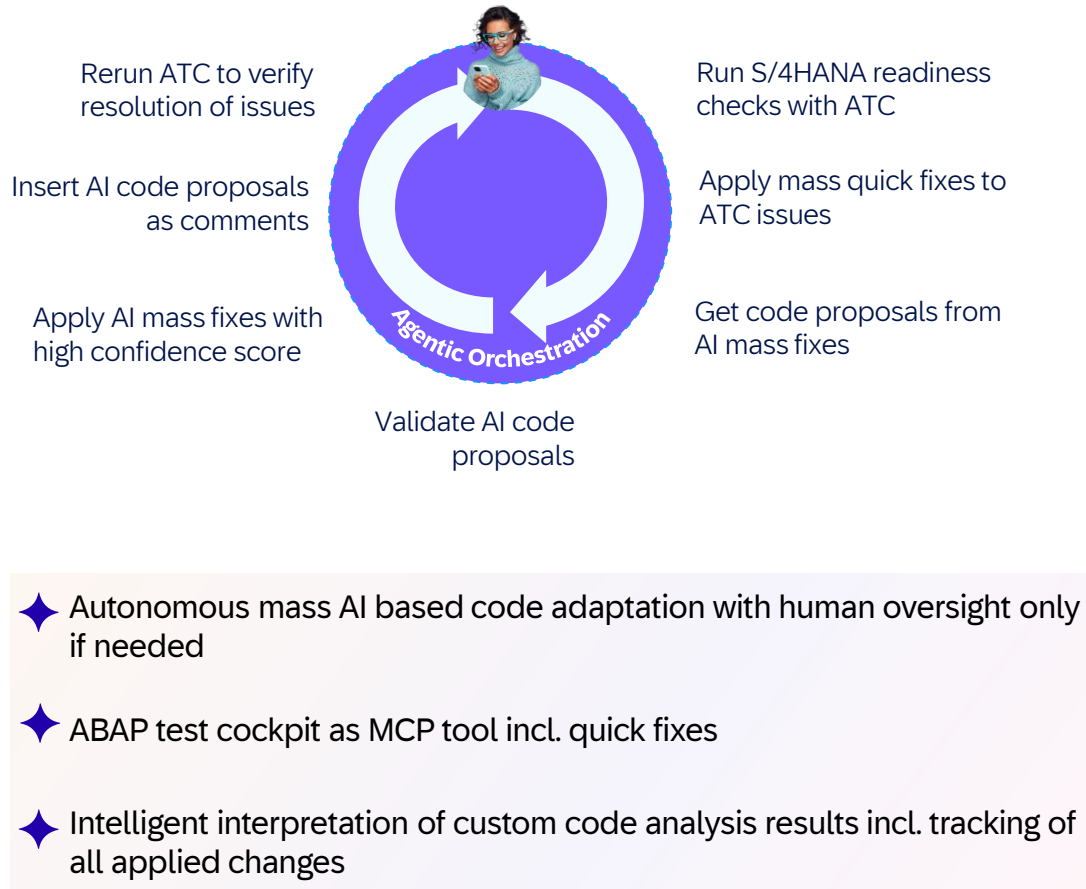
*only under the special RISE contract conditions announced at Sapphire keynote

- ABAP AI for lower releases of SAP S/4HANA Cloud Private Edition (2021–2025)
- ABAP development tools in Eclipse with ABAP MCP server and 3rd party AI assistant
- **Agentic Custom Code Migration:** Refactor custom code to accelerate ECC to S/4HANA transformation

SAP Joule for Developers, ABAP AI capabilities

Agentic workflow for custom code migration from SAP ECC to SAP S/4HANA Cloud Private Edition

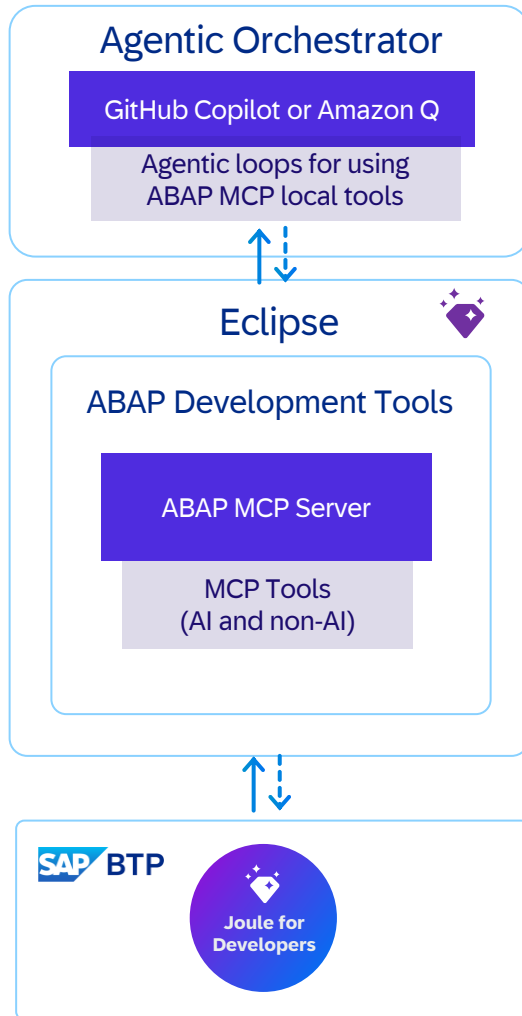
AI task “Migrate my legacy code to SAP S/4HANA compatible code”



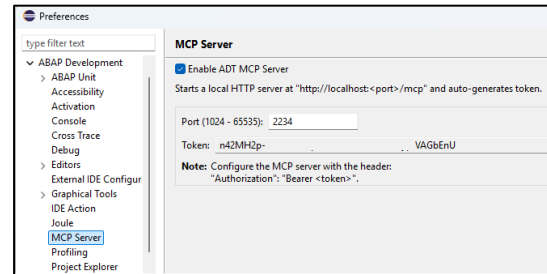
/ Check out [S/4HANA Custom Code Migration Agent - Accelerating ECC to SAP S/4HANA Transformation with Agentic AI](#)

How to prepare SAP S/4HANA Custom Code Agent to run

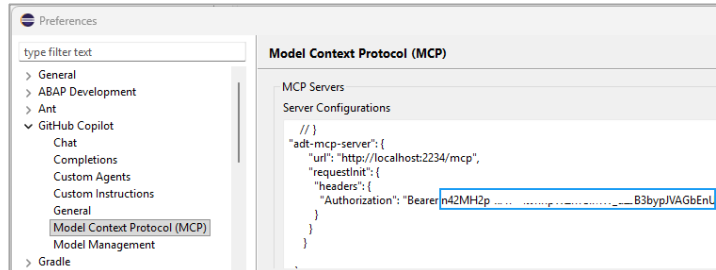
Prerequisites & Setup



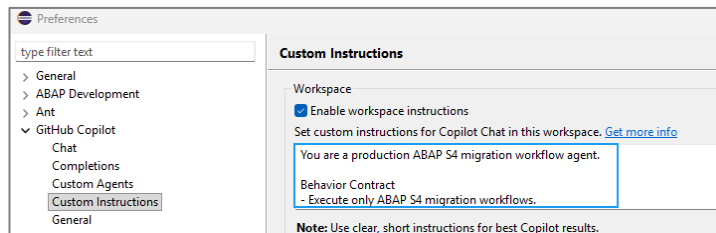
Enable MCP server



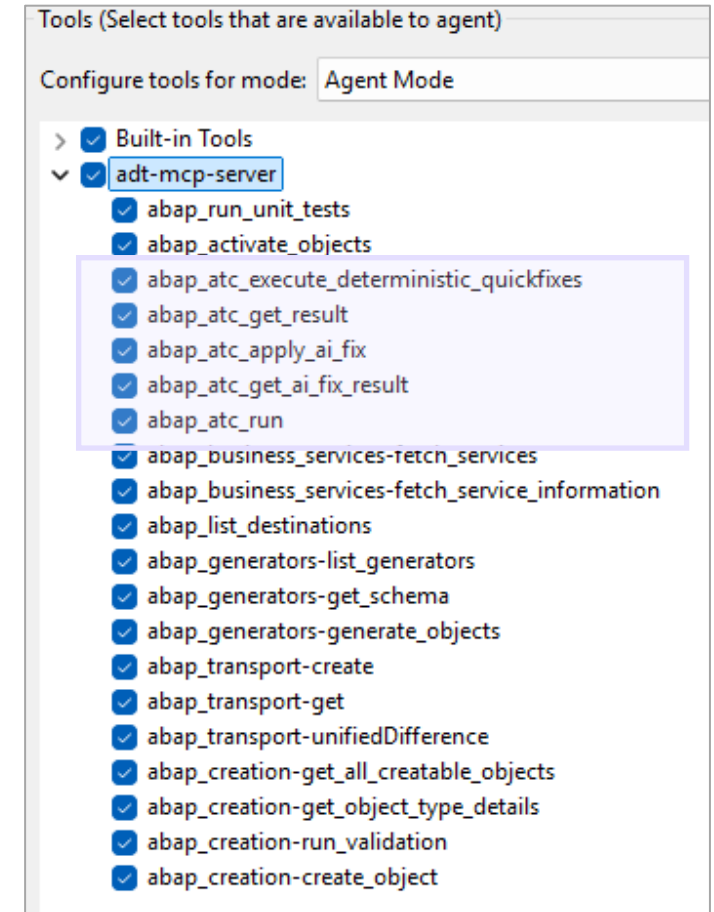
Configure MCP server



Provide custom instructions



ADT MCP tools incl. ATC tools



SAP S/4HANA Custom Code Migration Agent in action

Agentic workflow

Prompt agent to migrate your legacy package

Migrate package ZPGR_MM_PROCESSES_1 to SAP S/4HANA compliant code with variant S4HANA_READINESS_2025

Agent ▾ Claude Sonnet 4.6 - Medium ▾ ✂ ⌂

Run S/4HANA readiness checks

✦ The agent

- executes S/4HANA readiness checks
- applies deterministic quick fixes
- applies AI fixes with high confidence score
- adds remaining AI fixes as code comments

✦ All changes are tracked in the provided transport request

✦ Agent involves user for confirmation, action or decision

Migrati

status

GitHub Copilot

✓ Ran MCP tool: abap_atc_get_result - adt-mcp-server(MCP Server)

ATC Results

- **Input:**
- Worklist ID: 42010AEFAFF21FD19AC69F2DBC963FB
- Destination: A_...
- **Output:**
- Total Findings: **28** across **6** objects
- Errors: 12
- Warnings: 4
- Infos: 12
- Deterministic Quick Fixes Available: **14**
- AI Quick Fixes Available: **16**
- **Status:** [SUCCESS]

To proceed with applying the deterministic quick

Run deterministic quick fixes

GitHub Copilot

Transport request confirmed: **ACCK900683**. Proceeding with deterministic quick fixes

✓ Ran MCP tool: abap_atc_execute_deterministic_quickfixes - adt-mcp-server(MCP Server)

Deterministic Quick Fix

- **Output:**
- Status: Execution completed successfully
- Affected Objects (all activated):
- ZCL_D_MM_CHK_MAT_1 (CLAS/...
- ZDMM_MAT_CHANGE_1 (PROG/...
- ZSDS_SUPPL_CONF_1 (PROG/P...
- **Status:** [SUCCESS]

Run AI-based fixes

GitHub Copilot

Fetching AI fix status now.

✓ Ran MCP tool: abap_atc_get_ai_fix_result - adt-mcp-server(MCP Server)

AI Fix Status

- **Operation:** AI Fix Status
- **Output:**
- Status: ☒ Running
- Total: 14
- Processed: 8
- Skipped: 2
- **AI Fix Progress: 71% (processed: 8, skipped: 2)**
- **Status:** ☒ IN PROGRESS

AI fix is still running. Please trigger the chat again to do
"Check status" or similar to continue.

```
52 * Quick Fix Replace this statement by a SELECT statement with ORDER BY
53 * Transport: ACCK900683 demo
54 * Replaced Code:
55 * SELECT SINGLE matnr, werks
56 * FROM marc
57 * WHERE matnr = @ls_mara-matnr
58 * INTO @ls_marc.
59
60 * SELECT MATNR , WERKS
61 * FROM MARC WHERE MATNR = @LS_MARA-MATNR ORDER BY PRIMARY KEY
62 * INTO @LS_MARC
63 * UP TO 1 ROWS .
64 * ENDSELECT.
65 * End of Quick Fix
```

```
create public .
public section.
types:
* Begin of changes for S/4HANA compliance
* Legacy declaration (18-char material number) - commented out
* BEGIN OF ty_matnr,
* material TYPE bapi_mara-material,
* END OF ty_matnr .
* End of legacy declaration
BEGIN OF ty_matnr,
material TYPE mara-matnr, " Use 40-char MATNR domain to avoid length conflict
END OF ty_matnr .
types:
ty_matnr TYPE STANDARD TABLE OF ty_matnr WITH DEFAULT KEY .
* End of changes for S/4HANA compliance ty_matnr TYPE STANDARD TABLE OF ty_matnr

class-data GV_PRICE type P LENGTH 8 DECIMALS 2 value '1234.00' ##NO_TEXT.
class-data GV_DELTA type P LENGTH 8 DECIMALS 2 value '1.24' ##NO_TEXT.

class-methods CHECK_MATERIALS
importing
```

SAP S/4HANA Custom Code Migration Agent in action

Custom code migration results

Joule

Chat

Migration and ATC Analysis Results

GitHub Copilot

Fetching the AI fix status now.

✓ Ran MCP tool: abap_atc_get_ai_fix_result - adt-mcp-server(MCP Server)

AI Fix Status

• **Output:**

• Status: **Done** ✓

• Total: 14

• Processed: 12

• Skipped: 2

• **AI Fix Progress: 100% (processed: 12, skipped: 2, total: 14)**

• Syntax Errors Detected: 3

• Applied Changes by Confidence: High: 6, Medium: 4, Unreliable: 1

• **Status:** [SUCCESS]

Now generating the final CSV report.

✓ Created [42010AEFAFF21FD19ADE3B79988F44E2.csv](#)

Migration Complete ✓

• **File:** C:\Users\user\AppData\Local\Microsoft\GitHub\AI\42010AEFAFF21FD19ADE3B79988F44E2.csv

• **Contents:** Combined deterministic and AI fix results for package **ZPGR_MM_PROCESSES_1**

• 3 rows from Deterministic Quick Fix (objects activated): ZCL_D_MM_CHK_MAT_1, ZDMM_MAT_ZDSD_SUPPL_CONF_1

• 17 rows from AI Fix across 6 objects (12 processed, 2 skipped)

• ⚠ 2 findings skipped — "Functionality unavailable (no alternative planned)" for TYP CFX_B and TABLE_MATNR — require manual review (SAP Note 000224778)

• ⚠ 3 syntax errors detected in changed objects — manual review recommended

✦ Finally the agent provides detailed presentation about custom code migration result

1	Source	Check Title	Check Message	Object N	Obj	Sf	Referen	Referenced Object Name	Confidence Score
2	DeterministicFix			ZCL_D_MM_CHK_MAT_1	CLAS				
3	DeterministicFix			ZDMM_MAT_ZDSD_SUPPL_CONF_1	PROG				
4	DeterministicFix			ZDSD_SUPPL_CONF_1	PROG				
5	AI Fix	1 S/4HANA: Search for Usages of Simplified Objects	Functionality unavailable (no alternative planned)	ZCL_D_MM_CHK_MAT_1	CLAS	0002	TTYP	CFX_BI_TT_BAPI_MATRETURN2	
6	AI Fix	1 S/4HANA: Search for Usages of Simplified Objects	Functionality unavailable (no alternative planned)	ZCL_D_MM_CHK_MAT_1	CLAS	0002	TTYP	TABLE_MATNR	
7	AI Fix	1 S/4HANA: Field Length Extensions	Compare length conflict (DOMA MATNR, Note: 0)	ZCL_D_MM_CHK_MAT_1	CLAS	0002	DOMA	MATNR	High
8	AI Fix	1 S/4HANA: Field Length Extensions	Compare length conflict (DOMA MATNR, Note: 0)	ZCL_D_MM_CHK_MAT_1	CLAS	0002	DOMA	MATNR	High
9	AI Fix	1 S/4HANA: Field Length Extensions	Arithmetic type conflict (Type DMBTR, Note: 000)	ZCL_D_MM_CHK_MAT_1	CLAS	0002	DTBL	DMBTR	High
10	AI Fix	1 S/4HANA: Search for Usages of Simplified Objects	Syntactically incompatible change of existing ft.	ZCL_D_MM_CHK_MAT_1	CLAS	0002	FUNC	BAPI_MATERIAL_EXISTENCECHECK	High
11	AI Fix	2 S/4HANA: Field Length Extensions	Old Compare type conflict (Type WRBTR, Note: 0)	ZDMM_MAT_ZDSD_SUPPL_CONF_1	PROG	0002	DTBL	WRBTR	High
12	AI Fix	2 S/4HANA: Field Length Extensions	Old Compare type conflict (Type WRBTR, Note: 0)	ZDMM_MAT_ZDSD_SUPPL_CONF_1	PROG	0002	DTBL	WRBTR	High
13	AI Fix	2 S/4HANA: Field Length Extensions	WRITE (DOMA MATNR, Note: 0002215424)	ZDMM_MAT_ZDSD_SUPPL_CONF_1	PROG	0002	DOMA	MATNR	High
14	AI Fix	2 S/4HANA: Field Length Extensions	WRITE (DOMA MATNR, Note: 0002215424)	ZDMM_MAT_ZDSD_SUPPL_CONF_1	PROG	0002	DOMA	MATNR	High
15	AI Fix	2 S/4HANA: Field Length Extensions	WRITE (DOMA MATNR, Note: 0002215424)	ZDMM_MAT_ZDSD_SUPPL_CONF_1	PROG	0002	DOMA	MATNR	High
16	AI Fix	1 S/4HANA: Search for Usages of Simplified Objects	Syntactically incompatible change of existing ft.	ZCL_D_MM_CHK_MAT_1	CLAS	0002	FUNC	BAPI_MATERIAL_GET_ALL	Medium
17	AI Fix	1 S/4HANA: Search for Usages of Simplified Objects	Syntactically incompatible change of existing ft.	ZCL_D_MM_CHK_MAT_1	CLAS	0002	FUNC	BAPI_MATERIAL_GET_ALL	Medium
18	AI Fix	1 S/4HANA: Search for Usages of Simplified Objects	Syntactically incompatible change of existing ft.	ZCL_D_MM_CHK_MAT_1	CLAS	0002	FUNC	BAPI_MATERIAL_GET_ALL	Medium
19	AI Fix	1 S/4HANA: Search for Usages of Simplified Objects	Syntactically incompatible change of existing ft.	ZCL_D_MM_CHK_MAT_1	CLAS	0002	FUNC	BAPI_MATERIAL_GET_ALL	Medium
20	AI Fix	1 S/4HANA: Search for Database Operations	DB Operation SELECT found (BSEG, see Note(s):)	ZDMM_MAT_ZDSD_SUPPL_CONF_1	PROG	0002	TABL	BSEG	Medium
21	AI Fix	1 S/4HANA: Search for Usages of Simplified Objects	Syntactically incompatible change of existing ft.	ZDMM_MAT_ZDSD_SUPPL_CONF_1	PROG	0002	FUNC	BAPI_MATERIAL_SAVEDATA	Medium
22	AI Fix	1 S/4HANA: Search for Usages of Simplified Objects	Syntactically incompatible change of existing ft.	ZDMM_MAT_ZDSD_SUPPL_CONF_1	PROG	0002	FUNC	BAPI_MATERIAL_SAVEDATA	Medium
23	AI Fix	2 S/4HANA: Field Length Extensions	WRITE (DOMA MATNR, Note: 0002215424)	ZDMM_MAT_ZDSD_SUPPL_CONF_1	PROG	0002	DOMA	MATNR	Unreliable
24	AI Fix	2 S/4HANA: Field Length Extensions	WRITE (DOMA MATNR, Note: 0002215424)	ZDMM_MAT_ZDSD_SUPPL_CONF_1	PROG	0002	DOMA	MATNR	Unreliable

Thank you.

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